Solutions to Fourth Problem Sheet
on
Complex Analysis
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Topic: Complex Differentiability

"Complex analysis is the good twin and real analysis the evil one: beautiful formulas and elegant theorems seem to blossom spontaneously in the complex domain."
-Peter Henrici

"Real analysis is the battlefield; complex analysis is the paradise."
-Paul Halmos

"The miracle of complex differentiability is that from one simple condition — the Cauchy-Riemann equations — an entire universe of rigidity and beauty emerges."
-Lars Ahlfors

"In the complex domain, differentiability once implies differentiability infinitely many times. Nature is more generous than we deserve."
-Henri Cartan

"The requirement of complex differentiability is so rigid, so demanding, that functions satisfying it reveal the deepest secrets of mathematics."
-Carl Ludwig Siegel

"Riemann's concept of the complex derivative unified geometry, analysis and physics in one stroke, an act of mathematical genius that may never be surpassed."
-Felix Klein

Problem 1

Let $q : \mathbb{C} \to \mathbb{C}$ and let $w \in \mathbb{C}$. Show that $\lim_{z \to w} q(z) = L$ if and only if $\lim_{z \to 0} q(z + w) = L$. Deduce that for any point z_0 in the domain of a function f, f is differentiable at z_0 if and only if

$$\lim_{h \to 0} \frac{f(z_0 + h) - f(z_0)}{h}$$

exists. In this case $f'(z_0)$ also equals the latter limit.

Solution 1

Suppose that $\lim_{z \to w} q(z) = L$, then given any $\varepsilon > 0$ choose $\delta > 0$ to be such that $0 < |z - w| < \delta$ implies $|q(z) - L| < \varepsilon$, and hence if $0 < |z| < \delta$, then (equivalently) $0 < |(z + w) - w| < \delta$ which implies $|q(z + w) - L| < \varepsilon$. The conclusion now readily follows. Next, suppose that f is differentiable at z_0, meaning that the limit

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = L$$

exists in $\mathbb{C}$, then rename $z = h$ to get

$$L = \lim_{h \to z_0} \frac{f(h) - f(z_0)}{h - z_0}$$

and by applying the previous result, this holds if and only if (take $q(z) = [f(z) - f(z_0)]/(z - z_0), w = z_0$)

$$L = \lim_{h\to 0} \frac{f(h + z_0) - f(z_0)}{h + z_0 - z_0} = \lim_{h\to 0} \frac{f(z_0 + h) - f(z_0)}{h}.$$

Recall that we write $L = f'(z_0)$, so the proof is done. ✡

Problem 2

Show that $f(z)$ is holomorphic if and only if $(f(z^*))^*$ is holomorphic. Discuss whether the derivative of the conjugate of a function is the same as the conjugate of the derivative.

Solution 2

($\Rightarrow$) If f is holomorphic, then by the Cauchy-Riemann equations we have

$$\partial_x u(x, y) = \partial_y v(x, y) \quad \text{and} \quad \partial_y u(x, y) = -\partial_x v(x, y).$$

On the other hand, $f^*(z^*) = u(x, -y) - iv(x, -y)$ so that

$$\partial_x(\Re(f)) = \partial_x u(x, y), \quad \partial_y(\Re(f)) = -\partial_y u(x, y), \quad \partial_x(\Im(f)) = -\partial_x v(x, y), \quad \partial_y(\Im(f)) = \partial_y v(x, y),$$

whence the conclusion is immediate. The converse also follows from the previous equation. For the discussion, notice that $f(z) = z$ is entire and its derivative is 1 for all $z \in \mathbb{C}$, whilst $f(z)^* = z^*$ is differentiable nowhere, and in general, when a function is holomorphic, its conjugate is nowhere holomorphic, so we conclude that the question may not even have sense because the derivative of the conjugate may not even exist, the question of whether a function can be holomorphic as well as its complex conjugate will be answered in Problem 4 iii). ✡

Problem 3

Let f and g be two functions such that $f(z_0) = g(z_0) = 0$, both differentiable at z_0 with $g'(z_0) \neq 0$. Prove that

$$\lim_{z\to z_0} \frac{f(z)}{g(z)} = \frac{f'(z_0)}{g'(z_0)}.$$

The latter is known as the **L'Hospital's rule** for complex functions.

Solution 3

Suppose that $f(z_0) = g(z_0) = 0$, then

$$\lim_{z\to z_0} \frac{f(z)}{g(z)} = \lim_{z\to z_0} \frac{f(z) - f(z_0)}{g(z) - g(z_0)} = \lim_{z\to z_0} \frac{\frac{f(z)-f(z_0)}{z-z_0}}{\frac{g(z)-g(z_0)}{z-z_0}},$$

and assuming differentiability of f, g at z_0, the desired result now follows readily. ✡

Problem 4

We say that a function f is **antiholomorphic** if it is the complex conjugate of a holomorphic function.

i) Show that an anitholomorphic functions satisfies the **Anti-Cauchy-Riemann equations** (also known as the **conjugate Cauchy-Riemann equations**)

$$\partial_x u = -\partial_y v \quad \text{and} \quad \partial_y u = \partial_x v.$$

ii) Is the converse also true?

iii) Can a complex function be holomorphic and anitholomorphic at the same time? If so, determine which type of functions, and if not, prove it.

Solution 4

i) The proof is straightforward.

ii) Yes, if $f = u+iv$ satisfies the conjugate Cauchy-Riemann equations, then $g = u-iv \equiv \Re(g)+i\Im(g)$ is such that
$$\partial_x \Re(g) = \partial_y \Im(g) \quad \text{and} \quad \partial_y \Re(g) = -\partial_x \Im(g).$$

iii) The two conditions combined can be shown to imply that $\partial_y v = \partial_y u = \partial_x u = \partial_x v = 0$ so that the only functions satisfying both are the constant functions.

Problem 5

Let $\mathcal{V} = C^2(U)$ with $U \subseteq \mathbb{R}$ be the linear space of functions $f : U \to \mathbb{R}$ at least class C^2. We define the **Laplace operator** $\Delta : C^2(U) \to C(U)$ as
$$\Delta \doteq \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}.$$

We say that a function $f \in \mathcal{V}$ is **harmonic** if $f \in \ker \Delta$.

Show that the Cauchy-Riemann operators factorise the Laplace operator via
$$\Delta = 4 \cdot \partial_z \partial_{z^*},$$
and thus deduce that any holomorphic function is harmonic. Conclude that the sum of a holomorphic function with an anitholomorphic function yields a harmonic function. Finally, write $z = x+ict$ and hence deduce that the general solution to the one-dimensional wave equation is given by
$$u(x,t) = f(x+ct) + g(x-ct),$$
with f, g any two differentiable functions of their arguments.

Solution 5

Compute
$$\partial_z \partial_{z^*} = \frac{1}{2}(\partial_x - i\partial_y)\frac{1}{2}(\partial_x + i\partial_y) = \frac{1}{4}\left(\partial_x^2 + \partial_y^2 + i\partial_x\partial_y - i\partial_y\partial_x\right),$$
it is a well-known fact (Clairaut's theorem) that $[\partial_x, \partial_y] = 0$ on $C^2(U)$ so that
$$\partial_z \partial_{z^*} = \frac{1}{4}\Delta \iff \Delta = 4 \cdot \partial_z \partial_{z^*}.$$
Recall that f is holomorphic on U if and only if $\partial_{z^*} f = 0$ on U so that, in particular, if f is holomorphic, then $\partial_z \partial_{z^*} f = \partial_z 0 = 0$ and thus the conclusion easily follows from the factorisation of the Laplacian. An anitholomorphic function is annihilated by ∂_z so that it is also harmonic due to the factorisation. If we write $z = x + ict$ then the sum of holomorphic plus anitholomorphic can be written as
$$f(x+ict) + g(x-ict),$$
which is a hint that $f(x+ct) + g(x-ct)$ is harmonic, in fact, we can compute the partial derivatives and it is straightforward to see that
$$\frac{\partial^2 f}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2 f}{\partial c^2} = 0 \quad \text{and} \quad \frac{\partial^2 g}{\partial x^2} - \frac{1}{c^2}\frac{\partial^2 g}{\partial c^2} = 0.$$

Problem 6

By considering the gradients of the real and imaginary parts of a holomorphic function, show that the contour lines of the former and the latter intersect orthogonally. This result is represented graphically in Figure 1.

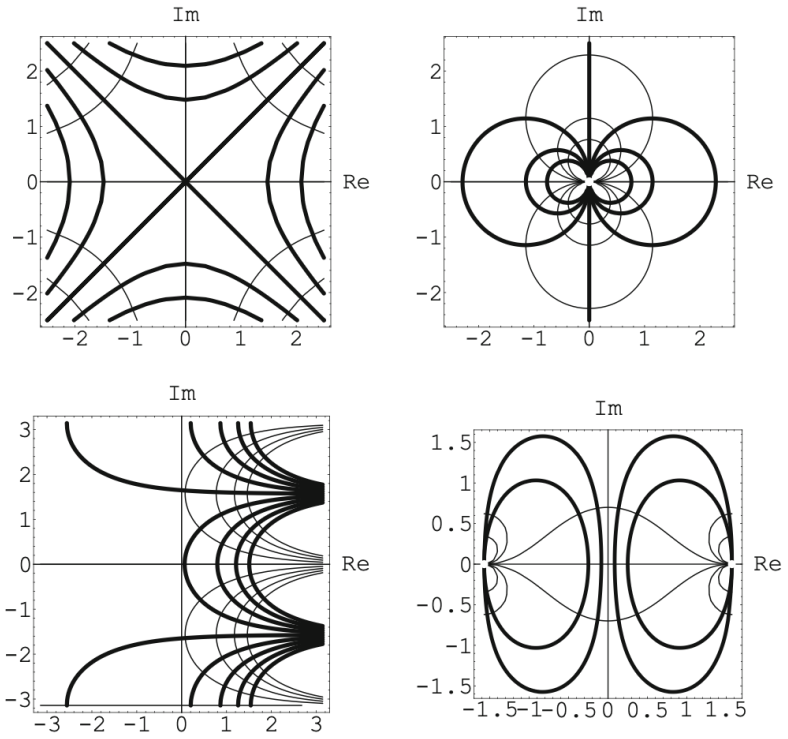


Figure 1: Orthogonal intersections of the real (thick trace) and imaginary (thin trace) parts of holomorphic functions z^2 (top left), $\frac{1}{z}$ (top right), e^z (bottom left), $\tan z$ (bottom right)

Solution 6

Let $f = u + iv$ and thus compute

$$\nabla u \cdot \nabla v = \partial_x u \partial_x v + \partial_y u \partial_y v,$$

which is 0 if u and v satisfy the Cauchy-Riemann equations. We conclude that the contour lines interset orthogonally at each point (the gradient of a function is orthogonal at each to the contour lines of such function). ✡

Problem 7

Let $z = x + iy$ and write x, y as functions of $z, \bar{z}$. Then deduce the Cauchy-Riemann operators by using the chain rule and treating $z, \bar{z}$ as independent variables. Verify that the Cauchy-Riemann operator $\partial_{\bar{z}}$ "differentiates with respect to $\bar{z}$ only" by showing that

i) $\partial_{\bar{z}} \bar{z}^n = n\bar{z}^{n-1}$ for $n \in \mathbb{N}$;
ii) $\partial_{\bar{z}}(\sin \bar{z} \cos z) = \cos \bar{z} \cos z$;
iii) $\partial_{\bar{z}}(z^2 \bar{z}) = z^2$;
iv) $\partial_{\bar{z}} e^{z\bar{z}} = z e^{z\bar{z}}$.

Solution 7

i) Straightforward, use the power rule for real differentiation.

ii) Compute $\frac{1}{2}\left(\partial_x + i\partial_y\right)\left(\sin\left(x-iy\right)\cos\left(x+iy\right)\right)$

$$= \frac{1}{2}(\cos\left(x-iy\right)\cos\left(x+iy\right) - \sin\left(x-iy\right)\sin\left(x+iy\right) + i[-i\cos\left(x-iy\right)\cos\left(x+iy\right) \\ -i\sin\left(x-iy\right)\sin\left(x+iy\right)]) \\ = \cos\left(x-iy\right)\cos\left(x+iy\right) = \cos\bar{z}\cos z.$$

iii) Straightforward.

iv) Use the properties of the exponential function and differentiate, the result follows directly.

Problem 8

Show that the function

$$f(z) = \begin{cases} z^5/|z|^4 & z \neq 0, \\ 0 & z = 0, \end{cases}$$

satisfies the Cauchy Riemann equations at $z = 0$, but at the same point it is not differentiable.

Solution 8

By definition, we have that (identify u, v via $f = u + iv$)

$$f(z) = \frac{(x+iy)^5}{(x^2+y^2)^2}$$

for $z \neq 0$, and $f(0) = 0$ so that

$$\partial_x u(0,0) = \lim_{h\to 0}\frac{u(h,0)-u(0,0)}{h} = \lim_{h\to 0}\frac{h^5/h^4}{h} = 1$$

and $\left(f(ih) = iy^5/y^4 \implies v(0,h) = h^5/h^4\right)$

$$\partial_y v(0,0) = \lim_{h\to 0}\frac{v(0,h)-v(0,0)}{h} = \lim_{h\to 0}\frac{v(0,h)-v(0,0)}{h} = \lim_{h\to 0}\frac{h^5/h^4}{h} = 1,$$

and hence $\partial_x u(0,0) = \partial_y v(0,0)$ and similarly, $\partial_y u(0,0) = -\partial_x v(0,0) = 0$. Nevertheless, taking the definition of complex derivative we see that

$$\lim_{w\to 0}\frac{f(w)-f(0)}{w} = \lim_{w\to 0}\frac{w^5}{|w|^4 w} = \lim_{w\to 0}\frac{w^4}{|w|^4}$$

which for w approaching from the real line yields 1 and approaching from the curve $y = x$ we have terms of the form $x + ix$ and hence the limit becomes

$$\lim_{w\to 0}\frac{x^4(1+i)^4}{4x^4} = \frac{(1+i)^4}{4} = -1.$$

We therefore conclude that f is not complex differentiable at $z = 0$.